

PHYSICS AS A WAY OF SAYING WHAT HAPPENS, THEN ASKING HOW IT CAN BE SO

Description before Explanation

MYP Physics Y10–11

A physicist can often describe a phenomenon before being able to explain it.

Observation → **Description** → **Explanation**

A **description** says what happens, what changes, what remains constant, or what pattern is observed. An **explanation** goes further. It introduces relations, interactions, constraints, models or mechanisms that make the observation intelligible.

In this worksheet, the explanatory question is usually:

How is it so?

That is different from asking “What is it for?” or “For what purpose does it happen?”

Physics usually does not explain natural phenomena by assigning them purposes.

1 Can you describe before you explain?

For each case below, separate what could be established directly from observation from what requires a physical explanation.

1. A falling ball

A ball is released from rest.

What could you observe directly?

Write a careful description without using the word *gravity*.

Now explain: how is it so that the ball speeds up?

2. A trolley after a push

A trolley is pushed and then released on a low-friction track. It continues moving.

Description only:

What tempting “explanation” should you avoid?

How is it so that the trolley continues to move after the push has ended?

3. A book on a table

A book remains at rest on a horizontal table.

Description only:

How is it so that the book remains at rest even though the Earth attracts it?

Which two systems must appear in a good explanation?

2 Statics: description can be richer than “it does not move”

4. A block on a slope

A block is placed on a slope. At one angle it remains still. At a larger angle it slides.

Give a description of the pattern.

What additional measurement could make the description more precise?

How is it so that the block can remain at rest on a slope at all?

5. Sliding or toppling?

Two blocks of different shapes are placed on the same adjustable slope. One slides first; the other topples first.

Describe the difference without explaining it.

Which observable dimensions might matter?

How is it so that one block topples while the other slides?

3 Thermal physics: an observed tendency before a mechanism

6. Hot block, cold water

A hot metal block is placed into cooler water.

What happens to the two measured temperatures?

Describe the final condition.

At this stage, can you state that thermal equilibrium has been reached even if you cannot yet explain the microscopic mechanism? Explain.

How is it so that the temperatures approach one another?

7. Heating without a temperature rise

Ice at its melting point receives energy while its temperature remains approximately constant.
What is the observation?

What is the description?

Why is “the energy has disappeared” not an explanation?

How is it so that energy enters while the temperature stays constant?

4 Waves: what travels, and what merely moves locally?

8. A pulse on a rope

A pulse travels along a stretched rope.

Describe what happens to the pulse.

Describe what happens to one marked point on the rope.

What important distinction can already be made before giving a full wave explanation?

How is it so that the disturbance travels while the rope itself does not travel with it?

9. Diffraction

Water waves pass through gaps of different widths.

Describe the pattern between gap width and spreading.

Could this description be correct even if you had not yet learned a full explanation of diffraction? Why?

What additional idea would begin to explain how the spreading occurs?

5 Fields: effects before hidden structure

10. The Earth's gravitational field

Release several different objects near the Earth's surface.

What common pattern can be described?

What does the description *not* yet tell you?

How is it so that an object accelerates downward without touching the Earth?

11. A compass near a bar magnet

Move a small compass around a bar magnet.

What can you observe?

Give a description that does not use the word *force*.

How is it so that the compass turns even though the magnet does not touch it?

12. Compare gravitational and magnetic fields

Complete the table from observations first. Add explanation only in the final column.

	Observation	Description	Beginning of explanation
Earth and falling object			
Bar magnet and compass			
Two bar magnets			

What can a magnetic field do that the gravitational field near Earth does not normally appear to do?

What similarity suggests that both are useful examples of a *field*?

6 Description, explanation, or merely renaming?

Label each statement:

D = description

E = explanation

R = renaming without explaining

1. The falling ball moves faster after each equal time interval. _____
2. The ball speeds up because it is falling. _____
3. The Earth exerts a gravitational force on the ball, producing downward acceleration. _____
4. The book remains at rest on the table. _____
5. The book stays still because it is in equilibrium. _____
6. The downward gravitational force is balanced by an upward contact force from the table. _____
7. The water and metal block eventually reach the same temperature. _____
8. They reach the same temperature because they reach thermal equilibrium. _____
9. Energy is transferred between the systems while a temperature difference exists, until there is no net transfer. _____
10. The wave spreads more when the gap is comparable with the wavelength. _____
11. The wave spreads because diffraction happens. _____
12. The compass rotates into a new orientation near the magnet. _____
13. The compass turns because of magnetism. _____
14. The magnet establishes a magnetic field, and the compass needle experiences a turning effect that aligns it with that field. _____

A statement can contain technical vocabulary and still fail to explain anything.

“It falls because of gravity”, “it spreads because of diffraction”, and “it stops changing because equilibrium is reached” may only rename the phenomenon.

A stronger explanation should show some physical relation:

system $\longleftrightarrow$ interaction $\longrightarrow$ change.

7 When is description enough?

Consider each case. Tick the statement you think is strongest and justify your choice.

A. Before learning Newton's laws

A student carefully measures a falling ball and concludes:

- ☐ “The ball falls because gravity pulls it down.”
- ☐ “The ball's downward velocity increases approximately uniformly during the fall.”

Which is scientifically stronger *at this stage*, and why?

B. Before learning a microscopic model of magnetism

A student maps the direction of a compass around a bar magnet.

- ☐ The student has done no physics until the origin of magnetism is explained.
- ☐ The student has established a reproducible spatial pattern even without a microscopic explanation.

Defend your answer.

C. Before learning a full theory of waves

A student discovers that narrower gaps produce more noticeable spreading.

Can this be a genuine scientific result before a full explanation is available?

8 From observation to explanation

For each phenomenon, complete all three stages.

Phenomenon	Observation	Description	How is it so?
Falling object			
Book on table			
Block on slope			
Hot block in water			
Pulse on rope			
Compass near magnet			

9 A more difficult question: what counts as explanation?

An explanation need not answer the deepest possible question.

At MYP level, saying that a magnet establishes a field which produces a turning effect on a compass can be a good explanation even though it does not explain why matter is magnetic at the quantum level.

Similarly, saying that the Earth exerts a gravitational force on an object can explain its acceleration even though it does not explain the ultimate nature of gravity.

Explanations therefore come in *levels*.

Choose one:

- gravitational field;
- magnetic field;
- thermal equilibrium;
- wave propagation.

Write:

A good MYP-level description:

A good MYP-level explanation:

A deeper question that remains unanswered:

10 Final reflection

Complete the following sentences.

1. A description can be scientifically valuable because ...
2. A description becomes an explanation when ...
3. Using a scientific word does not automatically explain a phenomenon because ...
4. One example in which observation came before explanation for me was ...

Teacher notes / possible responses

Falling ball

Description: after release, the ball's downward speed increases.

Explanation: the Earth exerts a gravitational force on the ball, giving a non-zero downward resultant force and hence downward acceleration.

Trolley

Description: after release, the trolley continues at approximately constant velocity.

Explanation: if friction and other resistive forces are negligible, the resultant force is approximately zero, so the acceleration is approximately zero. A continuing forward force is not required.

Book on table

Description: the book remains stationary.

Explanation: the Earth exerts a downward gravitational force while the table exerts an upward normal contact force. These balance, producing zero resultant force.

Thermal equilibrium

Description: the hotter body cools and the colder body warms until their measured temperatures become equal.

Explanation: energy is transferred between the systems while a temperature difference exists; at thermal equilibrium there is no net energy transfer.

Wave pulse

Description: the pulse travels along the rope while individual parts of the rope oscillate around approximately fixed positions.

Explanation: neighbouring parts of the rope interact, allowing the disturbance and energy to propagate while the material itself undergoes only local motion.

Magnetic field

Description: a compass needle changes orientation in a reproducible way as its position around the magnet changes.

Explanation: the magnet establishes a magnetic field in the surrounding region, and the compass needle experiences a torque tending to align it with the local field direction.

Gravity versus magnetism

Both give reproducible effects without direct contact, making a field description useful. Near Earth, gravity gives approx downward acceleration & is always attractive for ordinary matter. Magnetic interactions are strongly directional producing attraction/repulsion depending on orientation.

A useful oral prompt throughout the lesson is: *Could I have said this directly from careful observation?* If yes, it is probably description. Then ask:

What new physical relation has been introduced that makes the observation intelligible?

If none has been introduced, the supposed explanation may only be a relabelling.